DC/DH canopen 应用说明

Application description of DC / DH CANopen

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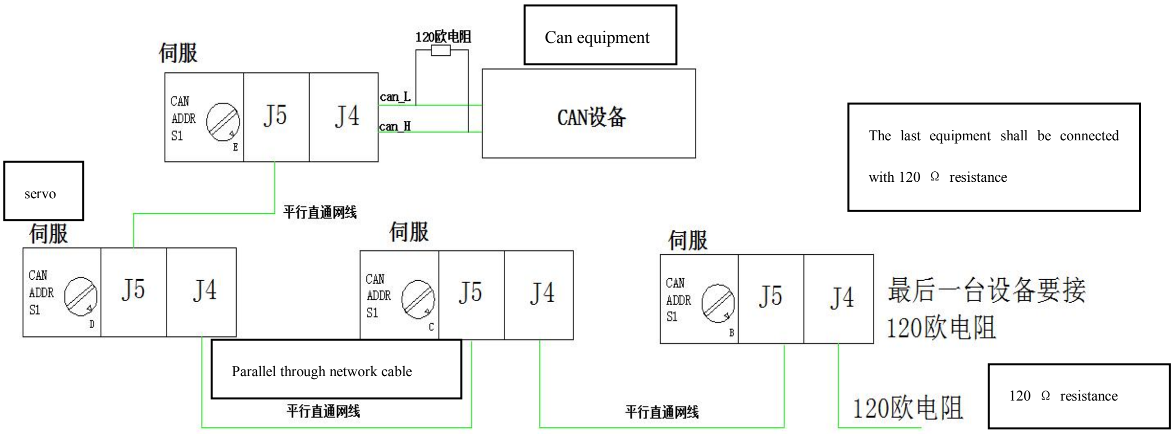
一．典型网络图 Typical network diagram

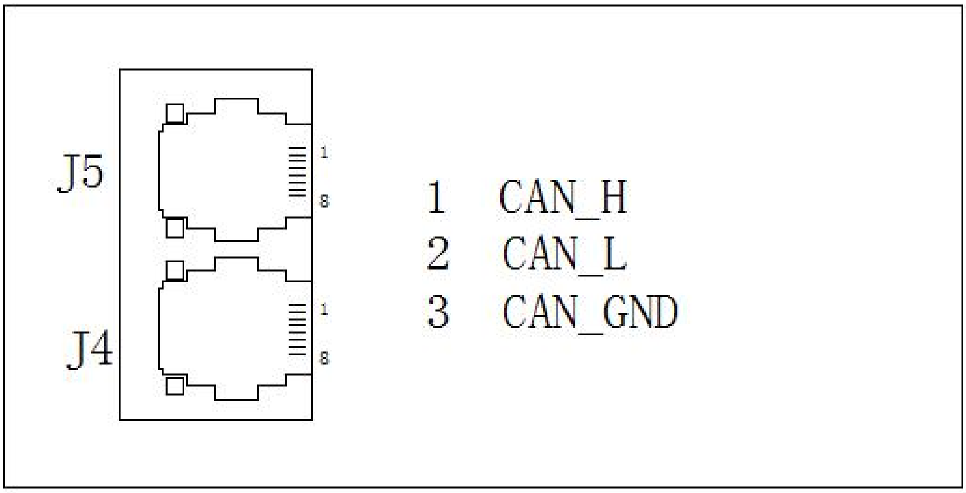
1.典型网络图接线 Typical network wiring diagram



注意：CAN网络上总的电阻是60欧，
如果CAN设备上有电阻的第一台驱
动器就不用加电阻

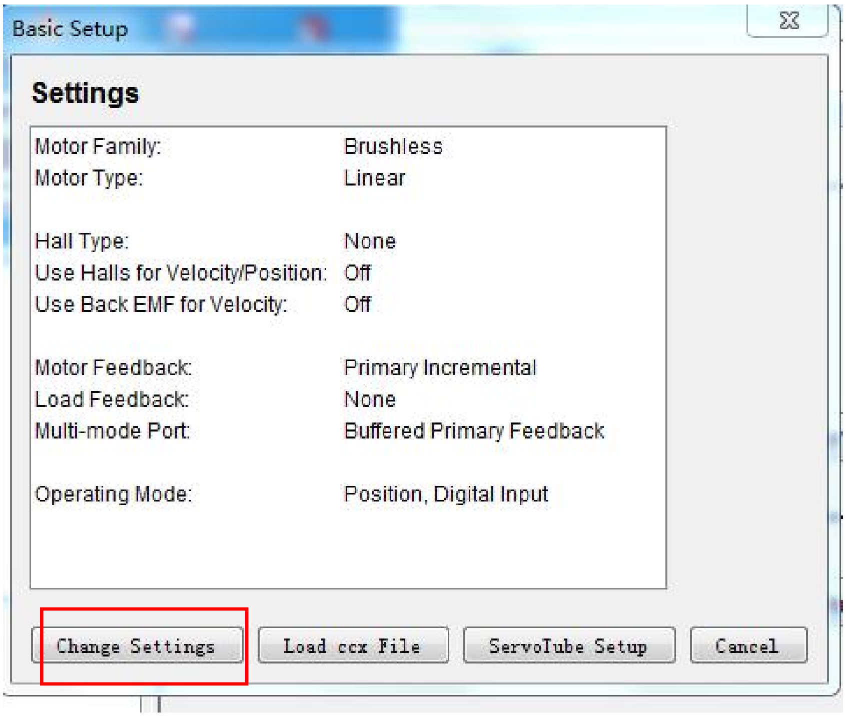
The total resistance on the can network is 60 Ω. If there is resistance on the can device, there is no need to connect the resistance on the first drive

2.驱动器 CAN 口定义 Driver can port definition

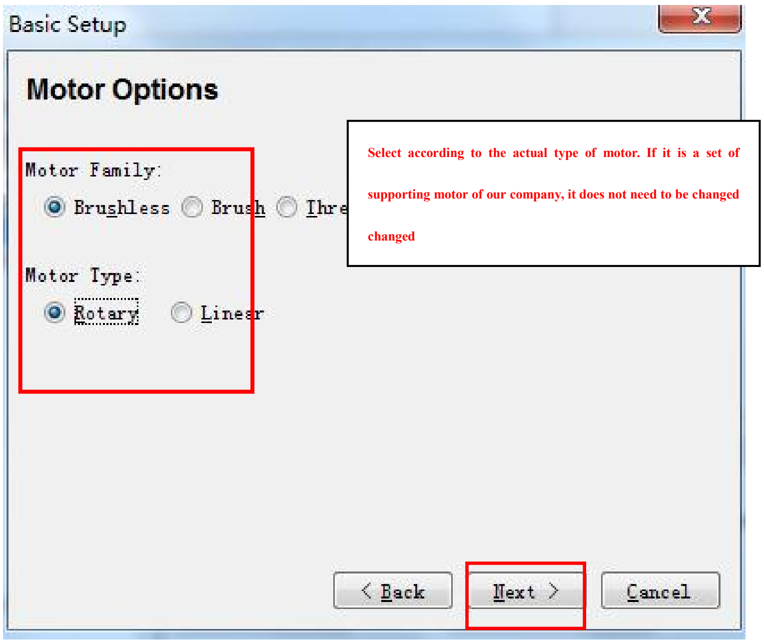


二．DCH 软件配置 Protocol command format

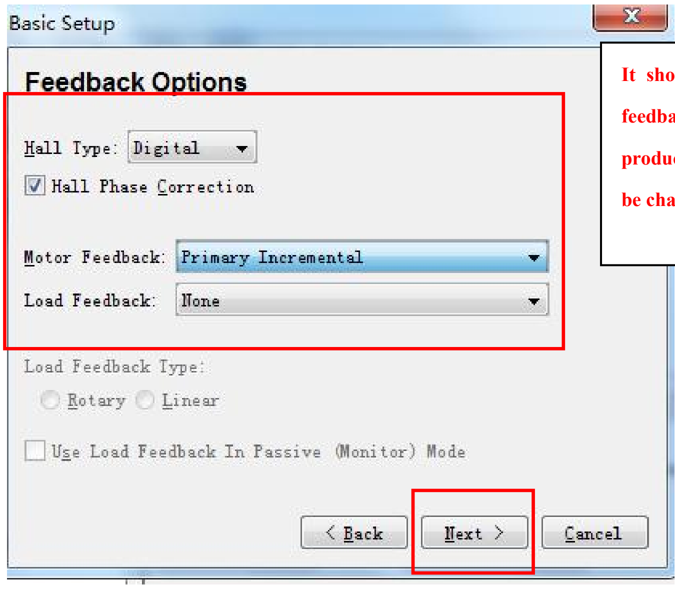
1. 启动软件,首先把驱动器设置成 CANOPEN 控制模式,按基本向导,按下列图进行配置,(如果我们配套电机和驱动器的直接下一步,到”操作模式选项”更改就可以了)
Start the software, first set the driver to CANopen control mode, and configure it according to the basic wizard and the following figure (if we go directly to the next step of matching the motor and driver, we can change it to "operation mode options")



选择”更改设置”Select change settings

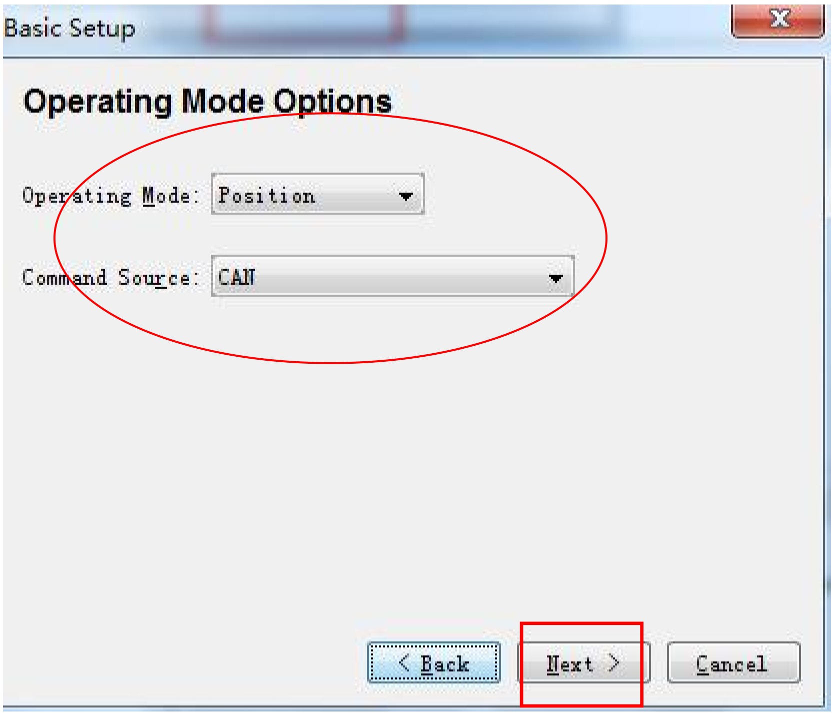


下一步”next

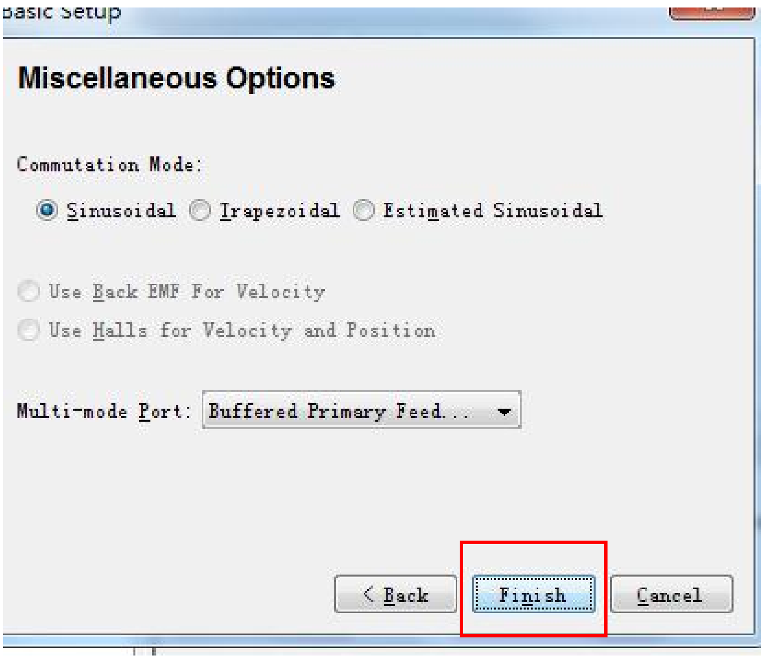


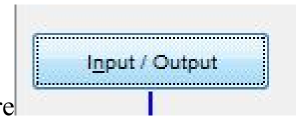
It should be set according to the motor feedback type you use. A set of supporting products of our company does not need to be changed

“下一步”next

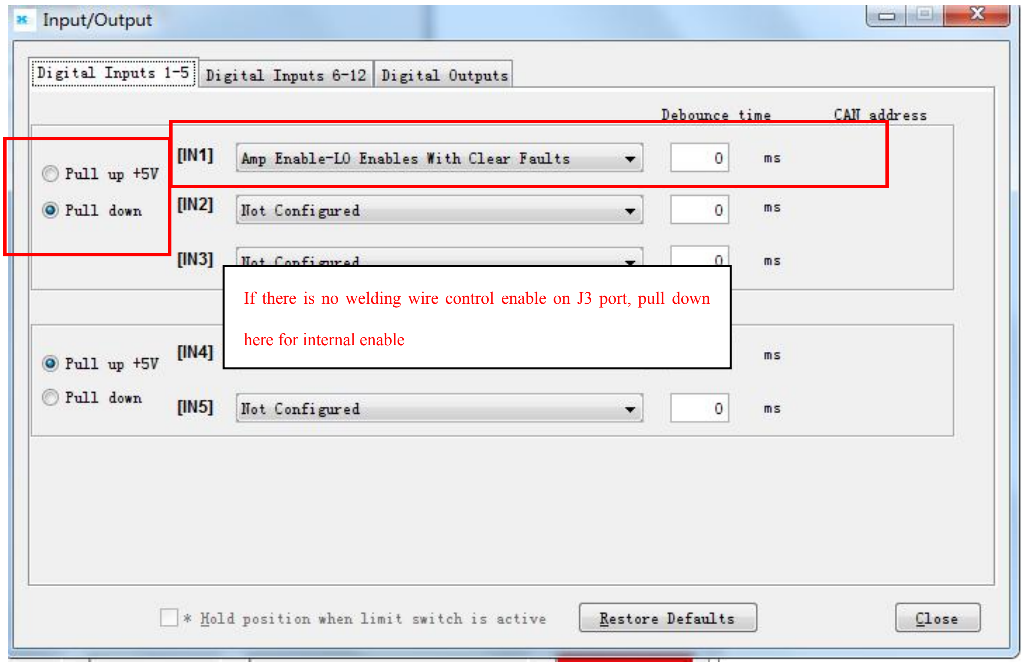


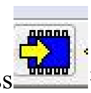
“下一步”next



软件主页面上 On the main page of the software 

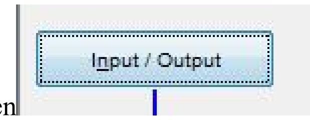
，进行驱动器使能配置 Drive enable



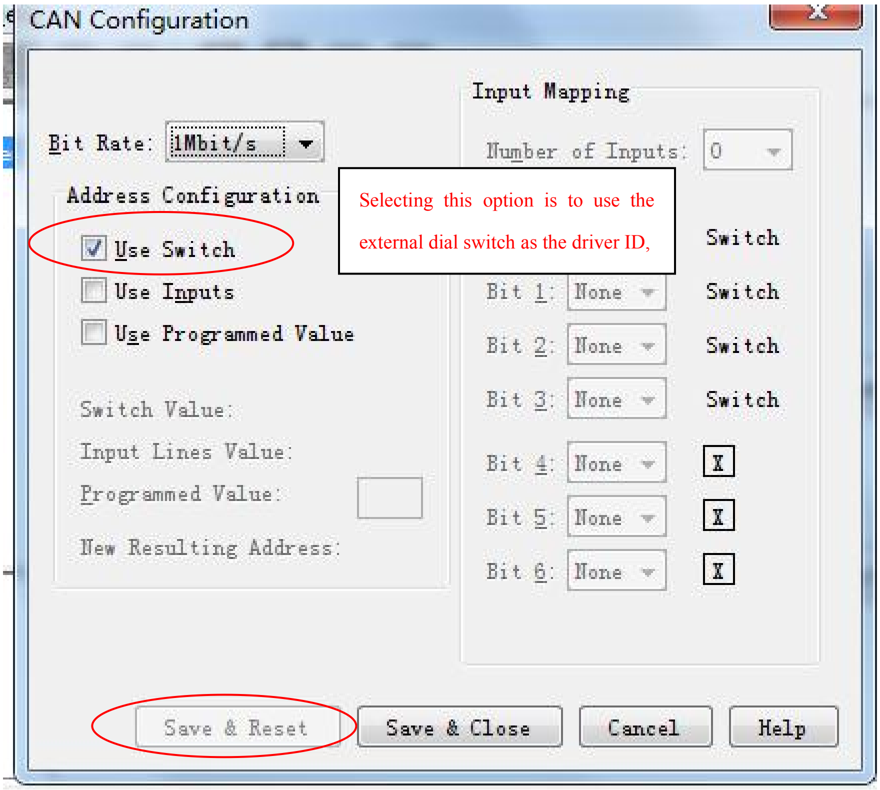
然后按完成,最后按 Then press finish, and finally press 把设置下载到 flash.这样就完成 CANOPEN 的配置了。Download the settings to flash This completes the configuration of CANopen.

驱动器波特率和站号设置

Drive baud rate and station number settings

打开 open

,如下图 As shown below



1. 选择”CAN”,选择波特率，然后选择 can 地址的给定方式，上图 can 地址是 1。最后按，“save & reset”。注意：上位机的波特率要跟这里的设定的一样。

Select "can", select baud rate, and then select the given mode of can address. The can address in the above figure is 1. Finally, press "save & reset". Note: the baud rate of the upper computer should be the same as that set here.

2. 这样一个 CAN 控制的驱动器设定就完成了。

In this way, the setting of a can controlled driver is completed

三. 命令格式 Command format

1. NMT 协议 NMT protocol

使节点进入 Operational 状态，发送命令：

NMT protocol enables the node to enter the operational state and send the command:

Cob-ID		data	
000		01	nodeID

使节点进入 Stop 状态，发送命令：

Make the node enter the stop state and send the command:

Cob-ID		data	
000		02	nodeID

使节点进入 Pre-operational 状态，发送命令：

Make the node enter the pre operational state and send the command:

Cob-ID		data	
000		80	nodeID

使节点进入 Reset-application 状态，发送命令：

Make the node enter the reset application state and send the command:

Cob-ID		data	
000		81	nodeID

使节点进入 Reset-communication 状态，发送命令：

Make the node enter the reset communication state and send the command:

Cob-ID		data	
000		82	nodeID

注：如果对所有节点发送命令，则 nodeID=0;

Note: if the command is sent to all nodes, nodeid = 0;

例：

如果使节点 0x06 进入 Operational 状态： 000 01 06

如果使所有节点进入 Pre-operational 状态：000 80 00

Example:

If node 0x06 enters operational status: 000 01 06

If all nodes enter the pre operational state: 000 80 00

2. node guard 协议 Node guard protocol

查询 CANOPEN 从站的状态，主站发送命令：

Query the status of CANopen slave station, and the master station sends the following command:

700+nodeID	

从站响应：Slave response:

700+nodeID	

t	state

t：toggle，0，1，0，1.....

State: 4=stopped

5=operational

7f=pre-operational

注：其中 t 值会 0，1 交替变化

Note: the value of T will change alternately between 0 and 1

3. heartbeat 协议 Heartbeat agreement

不需要主站发送请求命令，CANOPEN 从站周期性的发送其状态帧：

The master station does not need to send the request command, and the CANopen slave station periodically sends its status frame:

700+nodeID	

0	state

State: 4=stopped
5=operational
7f=pre-operational

4. SDO 协议 SDO protocol

命令格式 Command format

Identifier	Command	Index Low Byte	Index High Byte	Subindex	Data1	Data2	Data3	Data4

响应格式 Response format

Identifier	Command	Index Low Byte	Index High Byte	Subindex	Data1	Data2	Data3	Data4

读命令 Read command

主站发送命令：Master station sends command:

600+ NodeID	40	Index Low Byte	Index High Byte	Sub-index	00	00	00	00

从站响应： Slave response:

成功 success

Data length=1 byte

580+ NodeID	4F	Index Low Byte	Index High Byte	Sub index	d1	x	x	x

X：没有定义 No definition

Data length=2 byte

580+ NodeID	4B	Index Low Byte	Index High Byte	Sub index	d1	d0	x	x

X：没有定义 No definition

Data length=3 byte

580+ NodeID	47	Index low Byte	Index High Byte	Sub index	d2	d1	d0	x

X：没有定义 No definition

Data length=4 byte

580+ NodeID	43	Index low Byte	Index High Byte	Sub index	d3	d2	d1	d0

失败返回　　Failure Return

580+ NodeID	80	Index low Byte	Index High Byte	Sub index	SDO abort code error

写命令 Write command

主站发送命令：（要发送的原始数据 d0 d1 d2 d3）

Master station sending command: (original data to be sent d0 D1 D2 D3)

Data length=1 byte

600+ NodeID	2f	Index low Byte	Index High Byte	Sub index	d0	00	00	00

Data length=2 byte

600+ NodeID	2B	Index low Byte	Index High Byte	Sub index	d1	d0	00	00

Data length=3 byte

600+ NodeID	27	Index low Byte	Index High Byte	Sub index	d2	d1	d0	00

Data length=4 byte

600+ NodeID	23	Index low Byte	Index High Byte	Sub index	d3	d2	d1	d0

从站响应：Slave response

写入成功 Write successful

580+ NodeID	60	Index low Byte	Index High Byte	Sub index	00	00	00	00

写入失败 Write failed

580+ NodeID	80	Index low Byte	Index High Byte	Sub index	SDO abort code error

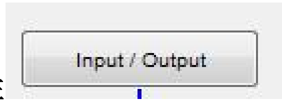
5. SYNC 协议 Sync protocol

80	00

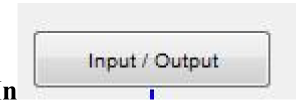
四. 电机回零设置 Motor zero return setting

本例子是外接开关当原点（开关接到 IN4 端子），开关信号是 NPN 输入

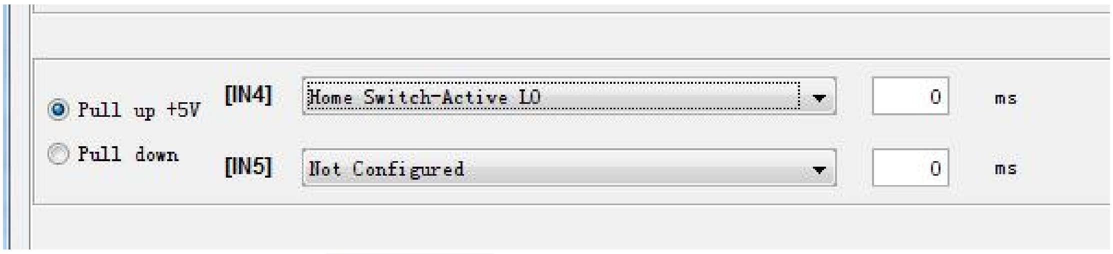
This example is an external switch. When the origin (the switch is connected to the in4 terminal), the switch signal is NPN input

1. 驱动器按照上述配置成 can 的控制模式，在 

中配置，IN4 为原点信号

The driver is configured into the control mode of can according to the above configuration,In 

configuration, in4 is the origin signal

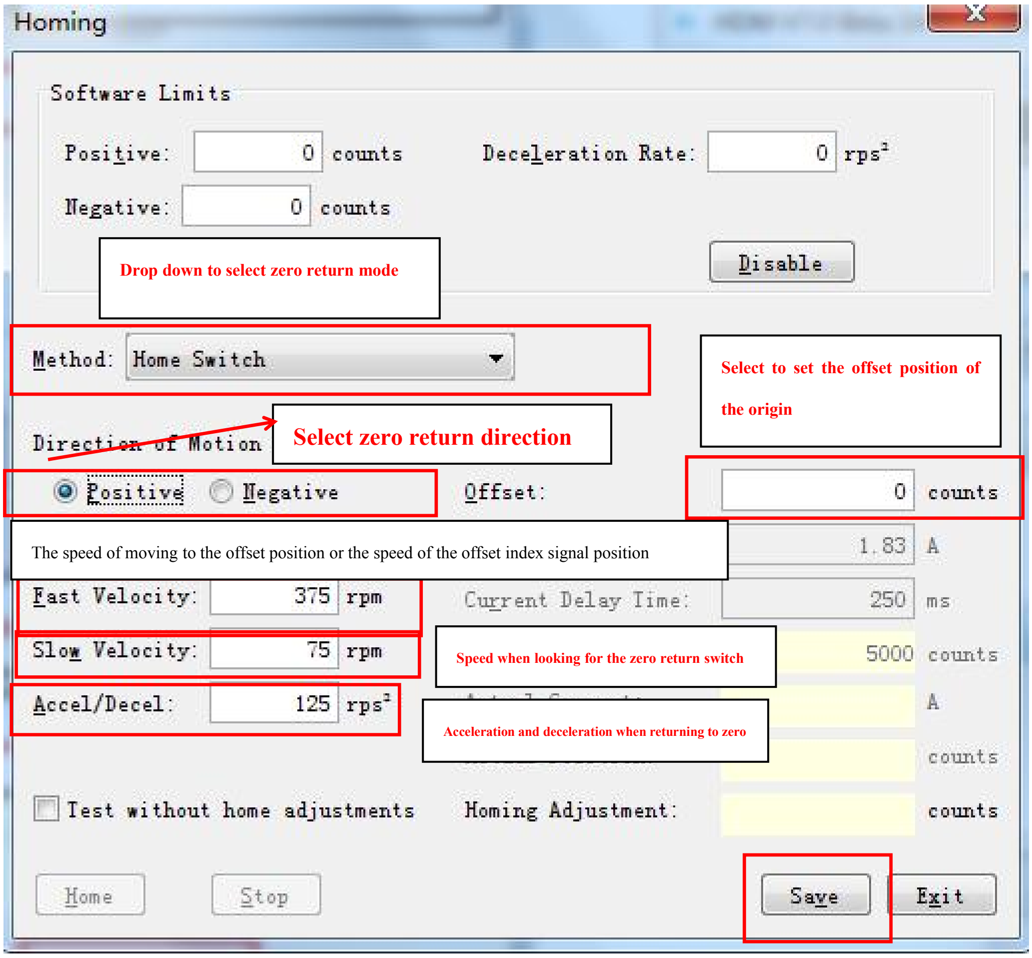


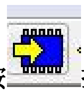
2. 在软件的主界面上点击 

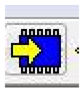
，设置回零的方式

Click on the main interface of the software 

,Set the way to return to zero 运动到偏置位置的速度或者偏置索引信号位置速度



3. 按"save"保存，最后在主界面按**把设置下载到 flash。**

Press "save" to save, and finally press in the main interface  **download the settings to flash**

4.进入 CAN 操作模式 Enter can operation mode

cob-id 0000 data 01 xx //（xx 是驱动器的节点，站号是 1 的就是 01，对所有节点操作就发 00）

(XX is the node of the drive. If the station number is 1, it is 01. 00 is sent for all node operations)

进入操作模式

Cob_id 000000 data 01 00

5.选择回零模式 Select zero return mode

Cob_id 60x Data 2F 60 60 00 06 00 00 00 //

6.使能驱动器 Enable driver

Cob_id 60x Data 2B 40 60 00 0F 00 00 00

7.启动回零 Start zero return

Cob_id 60x Data 2B 40 60 00 1F 00 00 00

注意:电机的回零方式还有多种，根据你系统要求可以在 DCH 软件上设置好（软件上的 HOME），然后发送指令就可以。

Note: there are many ways for the motor to return to zero. According to your system requirements, you can set it on the DCH software (home on the software) and then send instructions.

五、位置控制模式(电机反馈编码器为 2500 线)

Position control mode (2500 lines for motor feedback encoder)

先进入操作模式 cob-id 0000， data 01 xx（xx 是驱动器的节点，节点是 1 的就是 01，对所有节点操作就发 00）

First in operation mode cob ID 0000, data 01 XX (XX is the node of the drive, the node of 1 is 01, and 00 is sent for all node operations)

进入操作模式

Cob_id 000000 data 01 00

1、模式改成位置控制 Change mode to position control

Cob_id 60x Data 2F 60 60 00 01 00 00 00

2、位置模式下速度(0.1 count) ///速度地址是 0x6081

Speed in position mode (0.1 count) The speed address is 0x6081

Cob_id 60x Data 23 81 60 00 60 54 19 00 // 1000r/min

后 32 位数据为速度设定值（高位在后 ） 60 54 19 00 为 1000r/min）

分析：反馈编码器为 2500 线，驱动器内部进行了 4 倍频，这样电机每转对应 10000counts，驱动器的解析度是 0.1count/S.若希望设则速度为 1000RPM。则 10000/60 再乘以 10000 就等于 1666666，换成 16 进制就是 0x19546A.设置成 500RPM 则对应 833333.

The last 32-bit data is the speed setting value (the high bit is behind) 60 54 19 00 is 1000r / min)

Analysis: the feedback encoder is 2500 lines, and the frequency is quadrupled inside the driver. In this way, the motor corresponds to 10000 counts per revolution, and the resolution of the driver is 0.1count/s. If you want to set it, the speed is 1000rpm. Then 10000 / 60 multiplied by 10000 is equal to 1666666, which is 0x19546a in hexadecimal If it is set to 500 rpm, it corresponds to 833333

3、位置模式下加速度 (10 count)　　　　//////加速度地址 0x6083

Acceleration in position mode (10 count) //Acceleration address 0x6083

Cob_id　60x　　data　23 83 60 00 A0 86 01 00　　// 100r/s2

4、位置模式下减速度(10 count)　////减速度 0x6084

5、Deceleration in position mode (10 count) / / / deceleration 0x6084

Cob_id　60x　　data　23 84 60 00 A0 86 01 00　　//　100r/s2

5、设置目标位置 (1 count)　　//目标位置 0x607A

Set target location (1 count) / / target location 0x607a

Cob_id　60x　　data　23 7A 60 00 10 27 00 00　//10000count(电机转一圈)
Cob_id　60x　　data　23 7A 60 00 F0 D8 FF FF　　// -10000count 的位置

6、触发启动　　Trigger start

发送数据 2B 40 60 00 0F 00 00 00
发送数据 2B 40 60 00 5F 00 00 00
(这个是触发了相对位置,绝对位置是 1F,请参考 0X6040 的说明,附表一)
(this triggers the relative position and the absolute position is 1F. Please refer to the description of 0x6040 and attached table 1)

7.　读驱动器位置 Read drive location

Cob_id　60x　　data　40 64 60 00 00 00 00 00　(6064 电机位置)
（6064 motor position）

六.速度控制模式 Speed control mode

先进入操作模式 cob-id　0000 ，　data 01 xx（xx 是驱动器的节点，站号是 1 的就是 01，对所有节点操作就发 00）

First in operation mode cob ID 0000, data 01 XX (XX is the node of the drive, station number 1 is 01, and 00 is sent for all node operations)

1. 驱动器进入操作模式

Cob_id　000000　data 01 00

2. 把驱动器设置成速度模式发下列指令

Set the driver to speed mode and issue the following command

Cob_id 60x data 2F 60 60 00 03 00 00 00

3. 使能驱动器 Enable driver

Cob_id 60x data 2B 40 60 00 0F 00 00 00

4 写驱动器速度 (60ff 速度地址)

Write drive speed (60ff speed address)

Cob_id 60x data 23 FF 60 00 6A 6E 19 00 (1000r/min)
正转 Forward rotation

Cob_id 60x data 23 FF 60 00 96 91 e6 ff (-1000r/min)
反转 reversal

后 32 位数据为速度设定值（高位在后 ） 6A 6E 19 00 为 1000r/min）

分析：反馈编码器为 2500 线，驱动器内部进行了 4 倍频，这样电机每转对应 10000counts，驱动器的解析度是 0.1count/S.若希望设则速度为 1000RPM。则 10000/60 再乘以 10000 就等于 1666666，换成 16 进制就是 0x196E6A.设置成 500RPM 则对应 833333.

The last 32-bit data is the speed setting value (the high bit is behind) 60 54 19 00 is 1000r / min)

Analysis: the feedback encoder is 2500 lines, and the frequency is quadrupled inside the driver. In this way, the motor corresponds to 10000 counts per revolution, and the resolution of the driver is 0.1count/s. If you want to set it, the speed is 1000rpm. Then 10000 / 60 multiplied by 10000 is equal to 1666666, which is 0x19546a in hexadecimal If it is set to 500 rpm, it corresponds to 833333

4.断开使能 Disconnect enable

Cob_id 60x data 2B 40 60 00 00 00 00 00

5.读速度 Reading speed

Cob_id 60x data 40 69 60 00 00 00 00 00 (6069 速度地址)

七．力矩模式 Torque mode

先进入操作模式 cob-id 0000 ， data 01 xx（xx 是驱动器的节点，站号是 1 的就是 01，对所有节点操作就发 00）

First in operation mode cob ID 0000, data 01 XX (XX is the node of the drive, station number 1 is 01, and 00 is sent for all node operations)

1. Cob_id 60x data 2F 60 60 00 04 00 00 00 //选择力矩模式 Select torque mode
2. Cob_id 60x data 2B 40 60 00 0F 00 00 00

//使能驱动器 Enable driver

3. Cob_id 60x data 23 13 21 00 10 27 00 00 //电流上升速率,这里设置 10000mA/s

Current rise rate, MA / s. set 10000ma / s here

4. Cob_id 60x data 2B 40 23 00 55 00 00 00

//设置运行电流(单位 0.01A)Set the operating current (unit: 0.01A)

5. //221C 实际电流 221C actual current

Cob_id 60x data 40 1C 22 00 00 00 00 00

八．故障复位 Fault reset

Cob-id 0000 ， data 81 00

这个是把所有的节点复位了，单独把某个节点复位 data 就是发 81 0x(x 表示节点号)，这样要重新发送指令才可以运行。

This is to reset all nodes. To reset a node separately is to send 81 0x (X represents the node number), so that it can run only after sending instructions again.

九．节点超时,心跳上传 Node timeout, heartbeat upload

节点超时保护 Node timeout protection

发送 send out: cob-id 60x

data: 2B 0C 10 00 FA 00 00 00 //100C 是节点保护时间(单位:ms),这里设置 250

100C is the node protection time (unit: ms), and 250 is set here

发送 send out: cob-id 60x

data 2F 0D 10 00 04 00 00 00 //100D 是节点保护因子.这里设置 4

100D is the node protection factor Set 4 here

发送 cob-id 701 (远程帧) Send cob ID 701 (remote frame)

所以总的节点超时时间是 250*4=1000ms,如果在 1s 时间内没要收到远程帧,节点停止.

Therefore, the total node timeout is 250 * 4 = 1000ms. If the remote frame is not received within 1s, the node stops

心跳上传设置方法 Heartbeat upload setting method

发送：cob-id 60x

Data： 2B 17 10 00 FA 00 00 00 //1017 是心跳上传时间(单位:ms),这里设置 250。每隔 250ms 上传

1017 is the heartbeat upload time (unit: ms), and 250 is set here. Upload every 250ms

注意：心跳上传和节点保护只能使用一种，不能同时使用

Note: heartbeat upload and node protection can only be used in one way and cannot be used at the same time

说明：因为驱动器有默认的 PDO 映射，当驱动器状态变化时就会上传，上传的信息是

Note: because the drive has a default PDO mapping, it will be uploaded when the drive state changes. The uploaded information is

COB-ID 0x18X 数据为地址 0x6041 的内容

Cob-id 0x18x data is the content of address 0x6041

COB-ID 0x28X 数据为地址 （0x6041 + 0x6061）的内容

Cob-id 0x28x data is the content of address (0x6041 + 0x6061)

若不要接收 PDO 数据可以发送

If you don't want to receive PDO data, you can send it

cob_id 0x60x , data 2F 00 1A 00 00 00 00 00

关闭 0x18x 事件回复 Turn off 0x18x event reply

cob_id 0x60x , data 2F 01 1A 00 00 00 00 00

关闭 0x28x 事件回复 Turn off 0x28x event reply

十．PDO 映射例子

1A00

索引 Indexes	说明 explain		
0	映射到此 PDO 的对象总数。 可以将其设置为 0 以禁用 PDO 操作，并且必须在更改 PDO 映射之前将其设置为 0。 Describes the total number of objects mapped to this PDO. You can set it to 0 to disable PDO operations, and you must set it to 0 before changing the PDO mapping		
1	映射对象地址 Mapping object address		
2	映射对象地址 Mapping object address		
3			
4			

1800

索引 Indexes	说明 explain		
0			
1	设置 PDO,cob-id Set PDO, cob ID		
2	设置 PDO 传输方式 Set PDO transmission mode		
3			
4			

PDO 传输类型与 PDO 触发模式对应表

传输类型	PDO 触发条件（B=Both needed;O=One or Both）			PDO 传输
	SYNC	RTR	Event	
0	B		B	同步，非周期
1-240	O			同步，周期
241-251	保留			保留
252	B	B		同步，在 RTR 之后
253		O		异步，在 RTR 之后
254		O	O	异步，制造商特定事件
255		O	O	异步，设备子协议特定事件

上表中的 SYNC 代表接收到同步对象，RTR 代表接收到远程帧，Event 代表事件发生，如数值改变或定时时间到了等。B 代表两个触发条件均满足时触发 PDO 传输，而 O代表一个或者两个触发条件满足时均可触发 PDO 传输。

RPDO0 映射速度,

1. 601 23 00 14 01 01 02 00 80 //设置 PDO cob-id 201 红色字体是驱动器地址 ID Set up that the PDO cob-id 201 red font is the drive address ID
2. 601 2F 00 14 02 FF 00 00 00
 //传输方式,异步 Transmission mode, asynchronous
3. 601 2F 00 16 00 00 00 00 00 //断开 Disconnect PD01600
4. 601 23 00 16 01 20 00 FF 60
//速度 60ff 映射到 160001 Speed 60ff was mapped to 160,001
5. 601 2F 00 16 00 01 00 00 00 //启动 start PDO 1600
6. 601 23 00 14 01 01 02 00 00 //启动 start PDO 1400

1401 RPDO 映射成模式选择和使能 The RPDO is mapped into pattern selection and enabling

1. 601 23 01 14 01 01 03 00 80 //设置 PDO cob-id 红色字体是驱动器地址 ID Set up that the PDO cob-id red font is the drive address ID

2. 601 2F 01 14 02 FF 00 00 00
 //传输方式,异步 Transmission mode, asynchronous
3. 601 2F 01 16 00 00 00 00 00 ////断开 Disconnect PD01601
4. 601 23 01 16 01 08 00 60 60
 //模式 6060 映射到 160101 Mode 6060 maps to 160101
5. 601 23 01 16 02 10 00 40 60 ///6040 映射到 160102 6040 Mapped to 160102
6. 601 2F 01 16 00 02 00 00 00
 ///启动 PDO 1601 02 代表启动两个映射 Starting PDO 1601 02 represents starting both mappings
7. 601 23 01 14 01 01 03 00 00 //PDO cob-id 301

上面是 PDO 映射,映射完下面是发送 Above is the PDO mapping, after mapping below is the send

1. 301 03 0F 00 //发速度模式并使能,03 是模式,0f 是使能 Hair speed mode and enabling, 03 is mode, 0f is enabling
2. 201 00 00 14 00 //发送速度 Send speed

Tpdo,读数据

1、断开 pdo 0x1A0000 Disconnect the pdo 0x1A0000
cob_id 0x60x , data 2F 00 1A 00 00 00 00 00
2 、设置 0x1A0001 为速度反馈 Set 0x1A0001 for the speed feedback
cob_id 0x60x , data 23 00 1A 01 20 00 6C 60
3 、设置 0x1A0002 为驱动器状态 Set 0x1A0002 to the drive status
cob_id 0x60x , data 23 00 1A 02 10 00 41 60
4 、设置 0x180001 为 0x18x Set the 0x180001 to 0x18x
cob_id 0x60x , data 23 00 18 01 8**1** 01 00 00 //pdo cob_id 0x18**1**
5 、设置 0x180002 同步触发 Set the 0x180002 synchronization trigger
cob_id 0x60x , data 2F 00 18 02 01 00 00 00
//每个同步信号触发上传 Each synchronization signal triggers an upload
6 、使能 pdo 0x1A0000 Enable to enable pdo 0x1A0000
cob_id 0x60x , data 2F 00 1A 00 02 00 00 00
7 、发送同步信号(定时发送就会定时有数据回来)Send synchronization signal (with data back)
cob_id 0x80 , data 00
8.返回数据 Returns the data
Aa aa aa aa bb bb //前 4 个字节是速度值(aa),后俩个字节是驱动器状态信息 The first 4 bytes are the speed value (aa), and the last two bytes are the drive status information

十一. 保存 PDO 映射 Save the PDO map

映射完 PDO 可以发送下面子指令保存 PDO 的映射 After mapping the PDO, you can send the next face command to save the mapping of the PDO

DC 系列驱动器发送 DC Series Drive Send

1. Cob-id 60x, data 23 20 24 00 0A 00 00 00
2. Cob-id 60x, data 23 10 10 01 73 61 76 65

DE/DE2 系列驱动器发送 The DE / DE2-series drive is sent

3. Cob-id 60x, data 2B B3 21 00 00 01 00 00
4. Cob-id 60x, data 23 10 10 01 73 61 76 65

十二．附表 Schedule

附表一 Schedule I

控制字 Control word：0x6040

0	置 1 时，CAN 打开开关 When placing 1, the CAN turns on the switch
1	置 1 时，电压使能 When placing 1, the voltage enables
2	置 1 时，不急停 Set the 1 time, do not stop urgently
3	置 1 时，操作使能 When placing 1, the operation enables
4	0->1 指令执行 0-> 1, instruction executed
5	为 0 ，需要等待上个指令执行完成，为 1 立即执行 （配合 bit4） For 0, wait for the previous instruction to complete, for 1 immediately (with bit4)
6	0 、绝对模式 ，1 、相对模式 0. Absolute mode, 1. Relative mode
7	0->1 清除故障 0-> 1, clear the fault
8	指令执行中止 The execution of the instruction is suspended
9~15	保留 hold

附表二

状态字 status word：0x6041

0	置 1 时，CAN 允许被打开开关 At placing 1, the CAN is allowed to be switched on
1	置 1 时，CAN 开关打开 At position 1, the CAN switch is turned on
2	置 1 时，电机使能 ， 0 电机未使能 When placing 1, the motor is enabled, and 0 the motor does no
3	故障，当驱动器有严重故障时，该位置 1 Fault, position 1 when there is a serious drive failure
4	置 1 时，驱动器电压正常 When placing 1, the drive voltage is normal
5	置 0 时，电机急停 When placing 0, the motor stops sharply
6	置 1 时，CAN 进入停止模式 At position 1, the CAN enters the stop mode
7	置 1 时，驱动器有警告信息，但不影响运行 At 1, the drive has a warning message but does not affect operation
8	置 1 时，执行运动指令中止。At position 1, the motion command is aborted.

<table>
<tr><td>9</td><td colspan="4">置 1 时，远程操作，为 0 时执行内部函数
Set 1, perform the remote operation, and perform the internal function at 0</td></tr>
<tr><td>10</td><td colspan="4">置 1 时，运动指令执行完成
At position 1, the motion instruction execution is completed</td></tr>
<tr><td>11</td><td colspan="4">置 1 时，内部各控制环限制 When placing 1, each internal control ring is limited</td></tr>
<tr><td rowspan="3">12- 13</td><td colspan="2">位置模式 position mode</td><td colspan="2">回零模式 Back to zero mode</td></tr>
<tr><td>12</td><td>置 1 时，位置指令响应
At position 1, the position command responds</td><td>12</td><td>回零相应
Back to zero corresponding</td></tr>
<tr><td>13</td><td>置 1 时，位置跟随错误
When set 1, the position follows the error</td><td>13</td><td>回零错误
Back to zero error</td></tr>
<tr><td>14</td><td colspan="4">置 1 时，电机运行中，停止时清零
Set 1, the motor is running, zero clearance when stopped</td></tr>
<tr><td>15</td><td colspan="4">保留</td></tr>
</table>

附表三

0x6060　CAN 控制的模式选择 Pattern selection controlled by the CAN

值 value	说明 explain
1	位置模式 position mode
3	速度模式 Speed mode
4	力矩模式 Torque mode
6	回零模式 Back to zero mode

常用地址列表：

序号	CAN_INDEX (HEX)	长度 (字节)	操作方式	说明 explain
	100C	2	RW	CANopen 节点保护时间,配合 0x100D 使用,单位 (ms) CANopen node protection time, used with 0x100D, in unit (ms)
	100D	1	RW	CANopen 节点保护因子,和 0x100c 的乘积,为节点保护时间 The CANopen node protection factor, and the product of 0x100c, is for the node protection time
	6040	2	RW	控制字(附表一) Control words (Schedule I)
	6041	2	R	状态字(附表二) Status word (Schedule II)

	1002	4	R	驱动器状态寄存器 Drive status register
	2180	4	R	紧急状态寄存器 Emergency state register
	2181	4	R	事件锁存寄存器 Event lock-in register
	2184	4	R	限制状态掩码 Limit the status mask
	6060	1	RW	模式的控制模式选择(附表三) Control mode selection of the mode (Schedule III)
	6061	1	R	CAN 操作模式显示 The CAN operation mode is displayed
	2300	2	RW	操作模式　operator schema
	1001	1	R	
	2120	4	RW	跟随错误窗口 Follow the error window
	2182	4	R	故障掩码 Fault mask
	2183	4	R	锁存故障寄存器 Locch the fault register
	2310	4	RW	模式选择 mode selection
	6064	4	RW	电机实际位置(count) Motor actual position (count)
	6069	4	R	电机实际速度(0.1 counts/sec) Actual speed of the motor (0.1, counts / sec)
	221c	2	R	电机实际电流(0.01 amps) Actual current of the motor (0.01 amps)
	2200	2	R	模拟量输入 (mv) Analog quantity input (mv)
	2201	2	R	母线电压　(0.1 V) Bus line voltage (0.1 V)
	2202	2	R	驱动器温度 Drive temperature
	2190	2	R	端子输入状态 Terminal input status
	2191	2	RW	端子上下拉电平选择 Terminal up and down pull level selection
	2194	2	RW	端子输出状态与控制 Terminal Output Status and Control
	6067	4	RW	位置跟随窗口 Location follows the window

	6068	2	RW	位置跟随窗口时间 Location follows the window time
	60F4	4	R	位置错误 error in position
	60FB:1	2	RW	位置环比例(Pp) Position Ring Scale (Pp)
	60FB:2	2	RW	位置前馈() Position feedforward ()
	60FB:3	2	RW	速度前馈() Speed feedforward ()
	60FB:4	2	RW	增益倍数 Gain multiple
	607D:1	4	RW	软件负限位,回零后有效 Software negative limit, effective after return to zero
	607D:2	4	RW	软件正限位,回零后有效 The software is positive limited and valid after return to zero
	2253	4	RW	接近软件限制时的减速率 Reduced down rate when approaching the software limits
速度环				
	2100	4	RW	速度环最大加速度(1000 counts /s2) Maximum acceleration of the speed loop (1,000 counts / s2)
	2101	4	RW	速度环最大减速度(1000 counts /s2) Maximum reduction of speed loop (1000 counts / s2)
	2102	4	RW	速度环紧急减速度(1000 counts /s2) Speed ring emergency speed reduction (1,000 counts / s2)
	2103	4	RW	速度环最大速度(0.1 counts / sec)
	2104	4	RW	速度误差窗口 Speed error window
	606d	2	RW	
	2105	2	RW	速度错误窗口时间 Speed error window time
	2341	4	RW	命令速度(0x2300 设成 11 时的速度模式,速度值) Command Speed (0x2300 at 11, speed value)
	60F9:1	2	RW	速度环比例增益(Vp) Speed Ring Scale Gain (Vp)
	60F9:2	2	RW	速度环积分增益(Vi) Speed Ring Integral Gain (Vi)
	60F9:3	2	RW	
	60F9:4	2	RW	
	60F9:5	2	RW	
电流环参数				
	2110	2	RW	峰值电流 peak point current(0.01amps)
	2111	2	RW	持续电流 dynamic current(0.01amps)

	2112	2	RW	I2t 时间 I2t time (ms)
	2113	4	RW	命令电流增加率 Command the current increase rate (mA/sec)
	2340	2	RW	命令电流 Command current (0.01amps)
	60f6:1	2	RW	电流环比例增益 Current ring proportional gain(Cp)
	60f6:2	2	RW	电流环积分增益 Current loop integral gain(Ci)
	60f6:3	2	RW	电流偏移 Current offset
	6071	2	RW	目标扭矩 rated torque/1000
	6076	4	RW	额定扭矩 rating torque
函数发生器功能				
	2330	2	RW	函数发生器配置 The Function Generator configuration 8193:梯形波形 trapezoid waveform 2:正弦波形 Sine waveform
	2331	2	RW	函数发生器的频率 Frequency of the function generator(Hz)
	2332	4	RW	函数发生器的幅度 The amplitude of the function generator
	2333	2	RW	发生的占空比,方波有效 The duty cycle, square wave effective (0.1%)
回零模式参数 Back-to-zero mode parameters				
	6098	1	RW	回零模式选择(在软件配置) Back to zero mode selection (in the software configuration)
	6099:1	4	RW	回零高速速度 Back to zero high speed(0.1counts/sec)
	6099:2	4	RW	回零低速速度 Back to zero low speed(0.1counts/sec)
	609A	4	RW	回零时的加减速 Increase and deceleration back to zero(10counts/sec2)
	607C	4	RW	原点偏移位置 Origin offset position(counts)
	2351	2	RW	硬停止回零模式延时 Hard stop back to zero mode delay(ms)
	2350	2	RW	硬停止模式电流 Hard-stop-mode current(0.01A)
	2352	2	RW	

位置模式 position mode				
	2121	4	RW	S 曲线的加加速度 Acceleration of the S-curve(100counts/sec2)
	2252	2	R	轨迹曲线状态 Track curve state
	2122	4	R	轨迹生成器目标位置.脉冲输入时有用 Track Builder target location. Useful when using a pulse input
	607A	4	RW	给定目标位置 Given the target location
	6081	4	RW	轨迹位置模式生成器的速度 The Speed of the Track Position Mode Generator(0.1counts/sec)
	60FF	4	RW	轨迹速度模式生成器的速度 The speed of the Track speed pattern generator (0.1counts/sec)
	6083	4	RW	加速度 accelerated velocity(10counts/sec2)
	6084	4	RW	减速度 retarded velocity(10counts/sec2)
	6085	4	RW	急停减速度 Rapid stop reduction speed(10counts/sec2)
	6086	2	RW	轨迹曲线模式选择 Track curve pattern selection 0:是梯形曲线 It's a trapezoidal curve 3:是 s 形曲线 It's a s-shaped curve -1:只走速度 Just walk the speed
电机参数 parameter of electric machine				
	6410:2	2	RW	电机极数 Motor pole number
	6410:0B	4	RW	电机最高速度 Maximum speed of motor(rpm)
	6010:17	4	RW	电机编码器分辨率 Motor encoder resolution(counts)
驱动器属性 Drive properties				
	6510:3	2	R	驱动器峰值电流 Drive peak current
	6510:4	2	R	驱动器持续电流 Drive persistent current
	6510:5	2	R	I2t 时间 I2t time
	6510:6	2	R	驱动器最高电压 Maximum drive voltage
	6510:7	2	R	驱动器最低电压 Drive minimum voltage
	6510:9	2	R	驱动器最高温度 Maximum drive temperature

以上详细说明可以看 DCH_CANOPEN 文档

The above details can see the DCH_CANOPEN documentation

SDO 命令错误码 The SDO command error code

代码	说明 explain
0503 0000h	切换位未交替 The switch bit is not alternating
0504 0000h	SDO 协议超时 The SDO protocol was timed out
0504 0001h	客户端/服务器命令说明符无效或未知 The Client / server command specifier is invalid or unknown
0504 0002h	无效的块大小（仅块模式） Invalid block size (Block mode only)
0504 0003h	无效的序列号（仅块模式） Invalid serial number (Block mode only)
0504 0004h	CRC 错误（仅块模式） CRC Error (Block mode only)
0504 0005h	内存不足 run out of memory
0601 0000h	不支持访问对象 Access to the objects is not supported
0601 0001h	尝试读取只写对象 Try reading to a write-only objec
0601 0002h	尝试写入只读对象 Try to write to a read-only object
0602 0000h	对象字典中不存在对象 The object does not exist in the Object dictionary
0604 0041h	对象无法映射到 PDO The Object cannot have a possible mapping to the PDO
0604 0042h	要映射的对象的数量和长度将超过 PDO 长度 The number and length of the objects to be mapped will exceed the PDO length
0604 0043h	通用参数不兼容原因 Universal parameters are incompatible reasons
0604 0047h	设备中的一般内部不兼容 General internal incompatibility in the device
0606 0000h	由于硬件错误，访问失败 Access failed due to a hardware error

0607 0010h	数据类型不匹配，服务长度参数不匹配 Data type mismatch, service length parameter mismatch
0607 0012h	数据类型不匹配，服务参数长度太高 Data type does not match, the service parameter length is too high
0607 0013h	数据类型不匹配，服务长度参数太低 Data type does not match, the service length parameter is too low
0609 0011h	子索引不存在 The subindex does not exis
0609 0030h	超出参数的值范围（仅用于写访问） Out of the value range of the parameter (for write access only)
0609 0031h	参数值写得太高 The parameter values are written too high
0609 0032h	参数值写得太低 Parameter values are written too low
0609 0036h	最大值小于最小值 The maximum value is less than the minimum value
0800 0000h	一般错误 common fault
0800 0020h	无法将数据传输或存储到应用程序 Data could not transfer or stored to an application
0800 0021h	由于本地原因无法将数据传输或存储到应用程序控制 Data cannot be transferred or stored to the application control for local reasons
0800 0022h	无法将数据传输或存储到应用程序当前设备状态 Data could not transfer or stored to the application's current device state
0800 0023h	对象字典动态生成失败或没有对象字典存在 Object dictionary dynamic generation failed or no object dictionary exists
以上详细说明可以看 DS301_V4.02 文档 **Details above can see the DS301_V4.02 documentation**	